

WEGEN MHD

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#	\ __MHD__ /	#
#	WHAT IS IT? - Energy Generation based on Magneto-Hydrodynamics	#
#	SCOPE: - Local Power Infrastructure, Civil Scale Modular Installs	#
#	Contributing Electricity to National Power Grid	#
#	HOW? - Using extremely well known and well defined physics	#
#	known as Magneto-Hydrodynamics, which utilizes the	#
#	Lorentz Force and applying it to the worlds extremely under-	#
#	utilized dynamic coastal tidal zones and the ebb and flow of	#
#	large rivers.	#
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A saltwater Magnetohydrodynamic (MHD) generator converts kinetic energy directly into electrical power without moving mechanical parts (like turbines or armatures). It relies on Faraday's Law of Electromagnetic Induction: as conductive fluid (saltwater) flows through a channel which has a perpendicular (At right angles to the flow of liquid) magnetic field flowing around it, an electromotive force (EMF) is induced across the channel perpendicular to both the flow direction and the magnetic field vector. The conductor moving through this magnetic field generates an electrical current which can be used as a power source for any compatibly specified load (Such as a light bulb)

MHD generators are different from traditional electric generators in that they operate without moving parts (e.g. no turbines), so there is no limit on the upper temperature at which they can operate. Although high-temperature MHD Generators are out of the scope and focus of this whitepaper. RElevant though is the fact they have the highest known theoretical thermodynamic efficiency of any electrical generation method. This translates directly into higher generation efficiencies and longer lifetimes for electrical components used in an MHD system thanks to less thermodynamic wear and tear.

MHD dynamos are the complement of [MHD accelerators](#), which have been applied to pump [liquid metals](#), [seawater](#), and plasmas. Natural MHD dynamos are of great interest to the geophysics communities. The magnetic fields of the [Earth](#) and [Sun](#) are produced by these natural dynamos. There is a growing body of research and data across subjects that are highly interesting for a "Water-based Electrical Generation - MHD System". And thanks to the scalability and ubiquitous of electromagnetic waves what we learn here will be extremely valuable to new next-gen generation technologies that are not based on water but could be based on solar winds and other space based energy differential harvesting.

MAJOR SUCCESS IMPEDIMENTS:

1. The low [electrical conductivity](#) of [seawater](#)
2. Increasing [current density](#) is limited by [Joule heating](#) and water [electrolysis](#) in the vicinity of [electrodes](#)
3. increasing the magnetic field strength is limited by the cost, size and weight (as well as technological limitations) of [electromagnets](#)
4. The proposed seawater system currently requires power to generate power as there needs to be a current flowing across the anode and cathode

MAJOR REASONS FOR SUCCESS / TO SUCCEED:

1. Enough real world work has been done since the 1960's to ensure a broad set of solutions to apply to development hurdles as they appear
2. Overall the proposed system is exceedingly simple.
3. Thanks to the way electromagnetic fields work physically, if we can stabilize input to the system then there is 100% opportunity for free electrical energy generation
4. The design of such a MHD system for install on rocky outcrops of bays, in river mouths and hidden below ports and piers, will be completely out-of-sight and low-to-no environmental impact. Aligning nicely with environmental agencies guidelines.